Exercise Sheet 4

AD-AS Foundations, the Taylor Rule, and Oil Price Shocks

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Motivation. In 2022 European natural gas prices roughly tripled following Russia's invasion of Ukraine. Euro-area headline inflation peaked above 10 % in October 2022 while real GDP growth slowed sharply. The ECB faced a classic monetary-policy dilemma: raising rates to contain inflation risked deepening the output slowdown. This sheet builds the AD-AS model from scratch and uses it to analyse that episode quantitatively.

Notation. We write π^T for the central bank's *inflation target* and π^{eq} for *equilibrium inflation* (the value of π that solves the AD-AS system). These are different objects: π^T is a policy choice, π^{eq} is an endogenous outcome.

Parameters used throughout this sheet. Unless a question explicitly changes a parameter, use

$$\bar{A} = 2, \quad a = 1, \quad r^n = 2, \quad \pi^T = 2, \quad h = 0.75, \quad b = 0.5, \quad \gamma = 0.5, \quad \pi^e = 2.$$

Part I. The Taylor Rule in Action

1. The central bank sets the nominal interest rate according to

$$i = r^n + \pi + h(\pi - \pi^T) + b\,y,$$

where y denotes the output gap.

   (a) Compute the nominal interest rate when $\pi = 4$ and $y = -1$. What is the corresponding real interest rate?

   (b) Compute i when $\pi = 2$ and $y = 0$ (at target). Call this the *neutral nominal rate*.

   (c) Suppose energy prices push inflation to $\pi = 5$ while the output gap stays at $y = -1$. By how much does the CB raise i compared to the neutral rate?

   (d) The Taylor principle states that $h > 0$, i.e. the nominal rate rises *more than one-for-one* with inflation. Explain in two sentences why $h \leq 0$ would be destabilising.

2. Consider two central banks facing the same energy shock ($\pi = 5$, $y = -1$): Bank H uses $h = 1.5$ and Bank D uses $h = 0.25$.

   (a) Compute the nominal rate for each bank.

   (b) Which bank tolerates more inflation? Which bank inflicts a larger real-rate increase?

   (c) Without doing any algebra, sketch how the two Taylor rules look as lines in (i, π) space when the output gap is fixed at $y = 0$. Label the slopes and the intercept at $\pi = \pi^T$.

Part II. Deriving the AD Curve

Start from the IS curve

$$y = \bar{A} - a(i - \pi)$$

and the Taylor rule from Part I.

1. (a) Substitute the Taylor rule into the IS curve. Show that you obtain

$$y = \bar{A} - a\big[r^n + h(\pi - \pi^T) + by\big].$$

   (b) Collect all y terms on the left-hand side and solve for y as a function of π:

$$y = \frac{\bar{A} - ar^n}{1 + ab} - \frac{ah}{1 + ab}(\pi - \pi^T).$$

   This is the **AD curve**.

   (c) With the given parameters, verify that $\bar{A} - ar^n = 0$. What does this imply for the output gap when $\pi = \pi^T$?

   (d) Show that the AD slope is $\partial y / \partial \pi = -ah/(1 + ab)$. Compute the numerical value.

2. What shifts the AD curve (as opposed to a movement along it)?

   (a) Give the formula for how a fiscal expansion ($\bar{A} \to \bar{A} + \Delta$) shifts the curve. Compute the rightward shift when $\Delta = 0.75$.

   (b) Explain why an oil price rise does *not* shift the AD curve (holding r^n fixed), even though it raises costs for firms and households. What does it do instead?

   (c) Suppose the central bank raises its inflation target from $\pi^T = 2$ to $\pi^T = 3$. Show algebraically how this shifts the AD curve in (y, π) space.

3. (Comparison.) Compute the AD-implied output gap for $\pi \in \{1, 2, 3, 4, 5\}$ using the baseline parameters. Arrange your results in a small table. Explain in one sentence why the table illustrates the mechanism through which monetary policy stabilises demand.

Part III. Deriving the SRAS and LRAS Curves

1. **From the Phillips curve to SRAS.** Start from the expectations-augmented Phillips curve

$$\pi - \pi^e = -\alpha(u - \bar{u}) + s,$$

   where s is a *cost-push* or *supply-shock* term (positive s = adverse shock), and the Okun link $u - \bar{u} = -\lambda y$.

   (a) Show that substituting Okun into the Phillips curve gives

$$\pi = \pi^e + \gamma y + s, \qquad \gamma \equiv \alpha\lambda.$$

   (b) Explain in words why $\gamma > 0$: why does higher output raise inflation?

   (c) What does the supply shock $s > 0$ represent economically? Give one concrete example related to energy markets.

   (d) With $\pi^e = 2$, $\gamma = 0.5$, and $s = 0$, compute inflation for $y \in \{-2, -1, 0, 1, 2\}$. Verify that the SRAS curve slopes upward in (y, π) space.

2. **The Long-Run AS curve.** In the long run, expectations are fulfilled: $\pi = \pi^e$.

   (a) Starting from $\pi = \pi^e + \gamma y + s$, impose $\pi = \pi^e$. What must be true of y in long-run equilibrium (assuming $s = 0$)?

   (b) Draw a diagram with y on the horizontal axis and π on the vertical axis. Sketch the SRAS curve for $\pi^e = 2$ and the LRAS curve. Label the long-run equilibrium.

(c) Why is the LRAS curve vertical? What does this say about the ability of monetary policy to permanently raise output above potential?

Part IV. Finding Equilibria

Use the AD curve from Part II and the SRAS curve $\pi = \pi^e + 0.5\,y + s$ together.

1. **Baseline SRAS-AD equilibrium** ($s = 0$, $\pi^e = 2$).
   (a) Substitute the SRAS curve into the AD curve and solve for equilibrium output y^* and equilibrium inflation π^{eq}.
   (b) Verify that $(y^*, \pi^{eq}) = (0, 2)$.
   (c) Explain in words why equilibrium output equals potential output when there are no shocks and expectations are on target.
2. **SRAS-AD equilibrium after an energy shock** ($s = 1.5$, $\pi^e = 2$).
   (a) Repeat the algebraic steps from 1(a) with $s = 1.5$. Show that
   $$y^* = -0.6, \qquad \pi^{eq} = 3.2.$$
   (b) This outcome is called *stagflation*. Define the term and explain why the sign pattern (lower output, higher inflation) is a signature of a supply shock rather than a demand shock.
   (c) Draw the AD curve, the baseline SRAS (from question 1), and the shifted SRAS (with $s = 1.5$) in a (y, π) diagram. Mark both equilibria E_0 and E_1.
3. **Long-run equilibrium.**
   (a) Impose $y = 0$ on the AD curve and find the long-run equilibrium inflation rate. Is it equal to π^T? Explain why.
   (b) Now suppose the oil shock is *permanent* (it permanently raises production costs by $s = 1.5$). With $\gamma = 0.5$, compute where the LRAS shifts to. Find the long-run (y, π) pair if (i) the AD curve stays unchanged ($\pi^T = 2$), and (ii) the CB re-centres policy to keep $\pi^{eq} = \pi^T = 2$.
   (c) Monetary policy cannot permanently raise output above potential. Can it permanently lower inflation? Justify your answer using the model.
4. **De-anchored expectations.** Suppose the energy shock in question 2 causes firms and workers to revise expectations upward to $\pi^e = 3$ (while the shock itself subsides, so $s = 0$ again).
   (a) Solve for the new equilibrium (y^*, π^{eq}).
   (b) Compare to the original baseline from question 1. Explain why de-anchoring makes the inflation problem harder to solve even after the supply shock ends.
   (c) Can the CB immediately restore $(y, \pi) = (0, \pi^T)$? If not, what output gap is required to bring inflation back to π^T in the next period? Show the algebra and state the output cost.

Part V. The Policy Dilemma under an Energy Shock

Context. When Russia cut gas supplies in 2022, the ECB faced a choice: raise rates aggressively to contain inflation, or move cautiously to limit the output contraction. This part makes that dilemma precise.

Use the oil shock from Part IV ($s = 1.5$, $\pi^e = 2$). Compare two Taylor-rule parameters: **baseline** $h = 0.75$ and **hawkish** $h = 1.5$.

1. For each value of h:
   (a) Re-derive the AD curve slope $\partial y/\partial \pi$.
   (b) Solve for equilibrium (y^*, π^{eq}) with $s = 1.5$. (Show all steps for the hawkish case; the baseline is already done in Part IV.)
   (c) Summarise your results in a table with columns: h, AD slope $\partial y/\partial \pi$, y^*, π^{eq}.
2. Using your results:
   (a) Can the hawkish central bank fully stabilise inflation at $\pi^T = 2$? If not, what would be required?
   (b) At what cost (in terms of output) does the hawkish rule contain inflation compared to the baseline?
   (c) In a (y, π) diagram, draw the two AD curves and the shifted SRAS. Note that the AD curve with the higher $|\partial y/\partial \pi|$ appears *flatter* in the (y, π) diagram (where y is on the horizontal axis). Mark the three relevant equilibria: $E_0 = (0, 2)$, $E_1 = (-0.6, 3.2)$, $E_1^H = (-1, 3)$. Label the "policy trade-off corridor" along the shifted SRAS between E_1 and E_1^H.
3. **Discussion question.** The ECB argued in 2022 that the energy shock was "temporary" and raised rates aggressively anyway.
   (a) Using the model, explain why a temporary shock still justifies a rate response if expectations risk de-anchoring.
   (b) What additional information would you need to decide the optimal h for this shock? Name at least two factors not in the model.

Part VI. Python Exercises

Open the starter notebook `week04_student_start.ipynb`. It contains imports, the parameter dictionary, and function skeletons. Complete the `# TODO` sections as described below.

Exercise P1: Build the AD and SRAS toolkit.

(a) Complete `ad_output(pi, par)` that returns the output gap implied by the AD curve. Use the formula from Part II.

(b) Complete `sras_inflation(y, par)` that returns equilibrium inflation given the output gap. Your function must use `par['pi_e']` and `par['s']`.

(c) Complete `solve_sras_ad(par)` that solves the two-equation system analytically and returns `{'y_star': ..., 'pi_eq': ...}`.

(d) Verify: with $s = 0$ the function should return $(y^*, \pi^{eq}) = (0, 2)$; with $s = 1.5$ it should return $(-0.6, 3.2)$. Print a small check table.

Exercise P2: Three-curve diagram.

(a) Plot AD, SRAS (baseline, $s = 0$), and SRAS (oil shock, $s = 1.5$) on the same axes. Add a vertical line at $y = 0$ representing LRAS.

(b) Mark the two SRAS-AD equilibria as scatter points and label them E_0 and E_1.

(c) Add the hawkish AD curve ($h = 1.5$) to the same figure and mark equilibrium E_1^H. Confirm visually that the hawkish AD is *flatter* (less steep) in the (y, π) diagram.

(d) Write a three-sentence interpretation cell explaining what a student should take away from comparing E_1 and E_1^H.

Exercise P3: Simulating an oil shock path.

Calibrate a four-period shock path meant to capture the 2022 European energy crisis: $s = [1.5,\ 1.0,\ 0.5,\ 0.0]$ (shock fades over four quarters). Assume static expectations: $\pi_t^e = \pi_{t-1}$.

(a) Complete the function `simulate_oil_shock(par, h, s_path)` that iterates the AS-AD model forward period by period and returns a `DataFrame` with columns `period, s, pi_e, y, pi`.

(b) Run the simulation for both $h = 0.75$ and $h = 1.5$.

(c) Plot the output gap and inflation in a side-by-side two-panel figure. Show both Taylor rules on the same axes with different line styles.

(d) Write a two-sentence interpretation: which rule brings inflation back to target faster? What is the cost?

Exercise P4 (Extension, optional): Real data.

Using the FRED API (get a free key at fredaccount.stlouisfed.org and paste it into the starter notebook), download `PNGASEUUSDM` (European natural gas prices) and `CP0000EZ19M086NEST` (euro-area HICP).

(a) Plot the year-on-year change in gas prices and HICP inflation on a dual-axis figure from January 2020 to December 2023.

(b) Identify visually the quarter in which the gas price shock was largest.

(c) Overlay the model's simulated inflation path (from P3, $h = 0.75$) on the HICP panel. Comment on whether the model captures the timing and magnitude of the actual inflation episode.